**Task 1: Paper Summary**

**Title**: Gradient-Based Learning Applied to Document Recognition

**Authors**: Yann LeCun, Léon Bottou, Yoshua Bengio, Patrick Haffner

**Source**: Proceedings of the IEEE, 1998

**Link**:
[https://www.researchgate.net/publication/2985446_Gradient-Based_Learning_Applied_to_Document_Recognition](https://www.researchgate.net/publication/2985446_Gradient-Based_Learning_Applied_to_Document_Recognition)

I studied the "Gradient-Based Learning Applied to Document Recognition" by LeCun et al. as part of my work on identifying handwritten numbers. Many people think of this paper as a turning point in the development of neural network-based methods for pattern recognition, especially for applications that use visual input, such handwritten writing.

The writers talk about how Convolutional Neural Networks (CNNs) may be used to find patterns in documents. They make a strong argument for the benefits of CNNs in finding hierarchical features. They argue convincingly for the advantages of CNNs in extracting hierarchical features directly from raw pixel input, reducing the need for handcrafted features which were dominant in earlier approaches. This approach aligns directly with the objective of my project, where I aim to build a classifier that learns to recognize digits based on pixel-level data rather than pre-extracted features.

The paper introduces the **LeNet-5 architecture**, a multi-layer CNN that was trained on the MNIST dataset the same dataset I am using for my project. LeNet-5 features convolutional layers, subsampling layers (now called pooling), and fully connected layers, which together form an end-to-end trainable system. The authors demonstrate how backpropagation can be utilized to train such architectures efficiently, a principle that remains at the core of deep learning today.

One of the key insights I gained from the paper is how CNNs exploit the **spatial structure** in images. Local receptive fields, shared weights, and subsampling collectively enable the network to capture translation-invariant features. This concept deepened my understanding of why CNNs are ideal for visual pattern recognition tasks.

What stood out to me is the breadth of experimentation the authors conducted. They didn't limit their work to digits alone but extended the same methodology to complex tasks like bank check digit reading and Latin character recognition. Their system consistently outperformed traditional techniques like k-NN or linear classifiers, especially in noisy, real-world data — a challenge I anticipate dealing with in my own implementation as well.

Reading this paper has helped me better understand the theoretical grounding of my project. It has also helped me make judgments on the architecture of my prototype CNN classifier, especially when it comes to choosing layers and filter sizes. This study still has important information that is useful today, even if subsequent designs like ResNet and EfficientNet have expanded on LeNet's work. In conclusion, this paper has been instrumental in shaping both my understanding and my project direction. It not only reinforced the importance of end-to-end learning in pattern recognition but also offered practical design wisdom that I have already begun applying.